

Federal Survey Question Consolidation

AI-Assisted Analysis for Burden Reduction | DRAFT — Not validated for citation

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Disclaimer

 Warning

DRAFT — These materials are in development and have not been independently validated.

The views expressed are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce.

This is exploratory research assessing feasibility of AI-assisted survey analysis methods.

The Problem

- Federal surveys ask overlapping questions
- Respondent burden + data silos
- **This report:** Capstone analysis — which questions can actually consolidate?

Brief — already covered in Reports 1-2

Why It Matters

For Respondents: - Declining survey response rates + increasing costs - Overlapping questions create unnecessary burden - Consolidation reduces redundant questions

For Federal Agencies: - **Expert time is the scarce resource**, not expert capability
- Traditional approach: weeks/months of manual comparison - **Our goal:** Multiply expert effectiveness → faster burden reduction

Dual value proposition: (1) Burden reduction for respondents (2) Expert efficiency to achieve it faster

The Challenge

- Determining consolidation **requires expert judgment** (this doesn't change)
 - Traditional timeline: **weeks to months** for manual review of 1,598 pairs
 - Our approach: **AI handles pre-work, experts focus on highest-value decisions**
 - Result: Structured expert review table ready for validation
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What Experts Get

What AI Provides	What Experts Do
Pre-screened pairs with best matches	Validate match quality
Initial F1/F2/F3 classification	Confirm or override
Documented reasoning	Understand and critique
Confidence scores	Prioritize review effort
Barrier codes	Verify diagnosis

76% classified with high confidence → spot-check **24%** flagged for expert review → where judgment adds most value

Starting Point

- 7,400+ survey questions across 49 federal surveys
- Mapped to Census Bureau standard taxonomy
- [Report 1 established concept classification]

Concept Classification (Report 1)

- LLM ensemble with arbitrator pattern
- Each question → best concept match
- Taxonomy: Demographics, Economics, Education, Health, etc.



Key Insight

! Important

Same concept swappable data

- “Income” questions can measure completely different constructs
- Demographics (age, sex, race) = expected overlap (minority of questions)
- **Specialized content** = where real analysis matters

Harmonization Framework

Adopted standard survey harmonization classification:

Level	Definition
F1	Directly consolidable — no transformation needed
F2	Consolidable with transformation — needs recoding
F3	Not consolidable — fundamental barriers

Level	Definition
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Barrier codes: CC (construct/concept), TC (temporal), RS (response scale)

Pairwise Comparison

- Within concept categories, compare all question pairs
- **1,598 pairs** analyzed (CPS-ACS, FoodAPS-ACS)
- Naive but exhaustive — expect many F3, looking for matches

Multi-Model Ensemble

Three frontier models to reduce bias:

Model	Behavior
OpenAI	Moderate synthesis, slight pro-self bias
Anthropic	High synthesis, neutral
Google	Deferential, conservative

Different profiles = documented finding, not noise

Agreement & Arbitration

- = **0.845** — high inter-rater agreement
- Voting, entropy, Borda scoring
- Arbitration resolves disagreements → final verdict

Evaluating AI for Survey Harmonization

Construct Validity: - Three different architectures (OpenAI, Anthropic, Google) with different training - **High convergence (= 0.845)** = task is well-defined, not model artifact

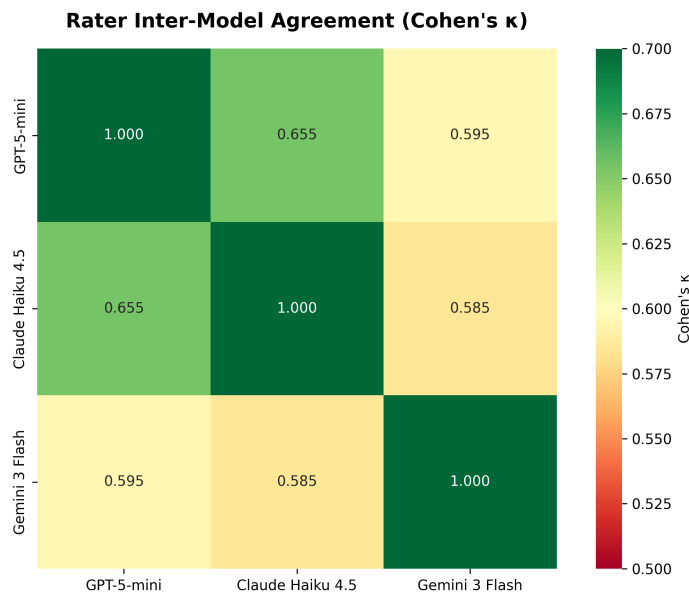
Documented Behavioral Differences: - Google: Deferential, conservative (low synthesis) - OpenAI: Moderate synthesis, slight self-bias - Anthropic: High synthesis, neutral

Cost/Quality Design: Fast models rate → flagship models arbitrate

Result: Reusable methodology for future survey harmonization work

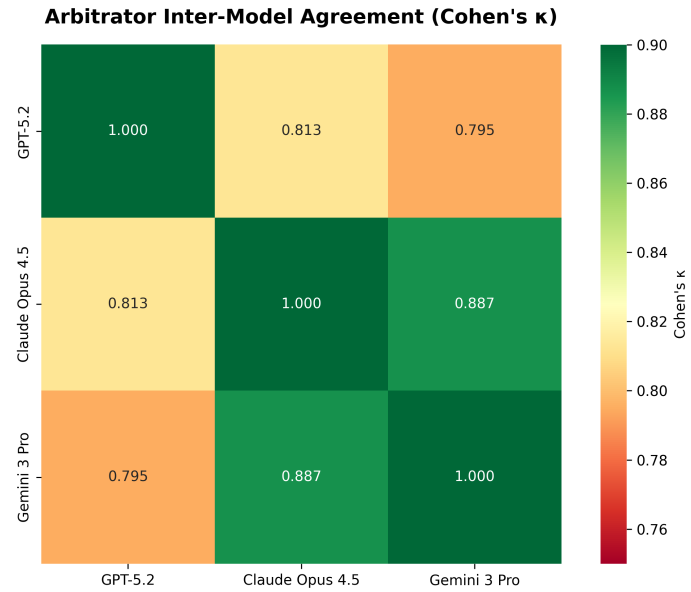
Construct Validity: Why Trust These Results?

Three independent architectures → same judgments = well-defined task



Rater Agreement

Fleiss = 0.611 (substantial)



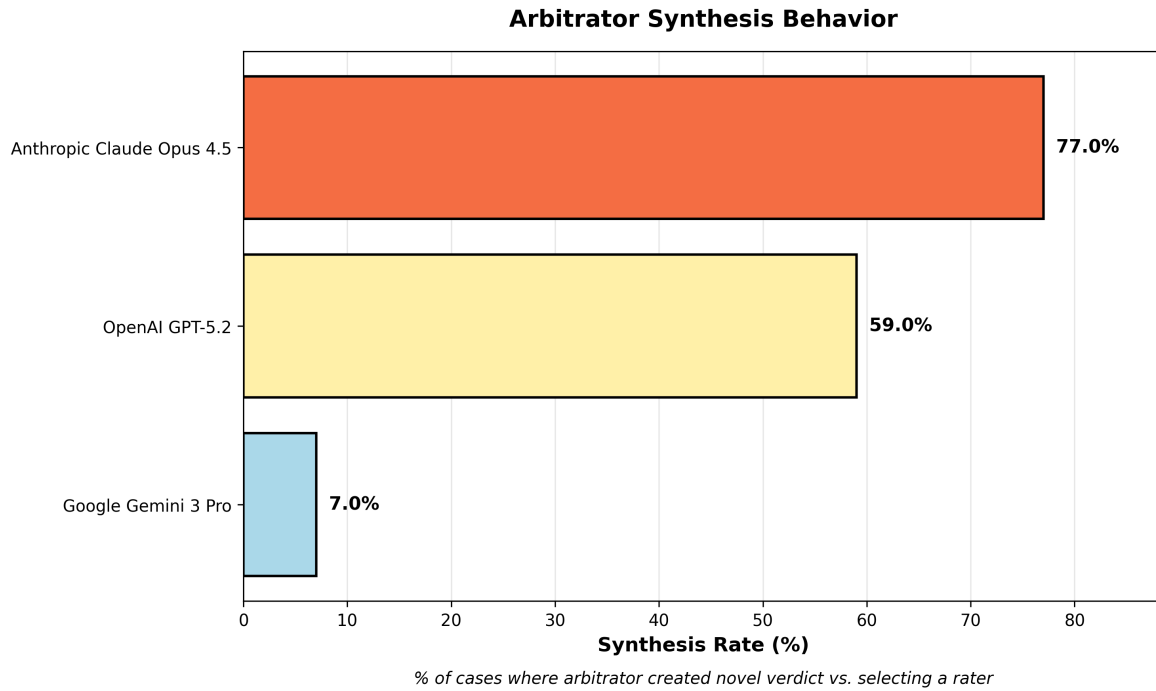
Arbitrator Agreement

Fleiss $\kappa = 0.843$ (almost perfect)

Convergence INCREASES with reasoning depth ($0.611 \rightarrow 0.843$)

Arbitrator Behavioral Differences

Not a bug — a feature. Diversity captures multiple valid interpretations.

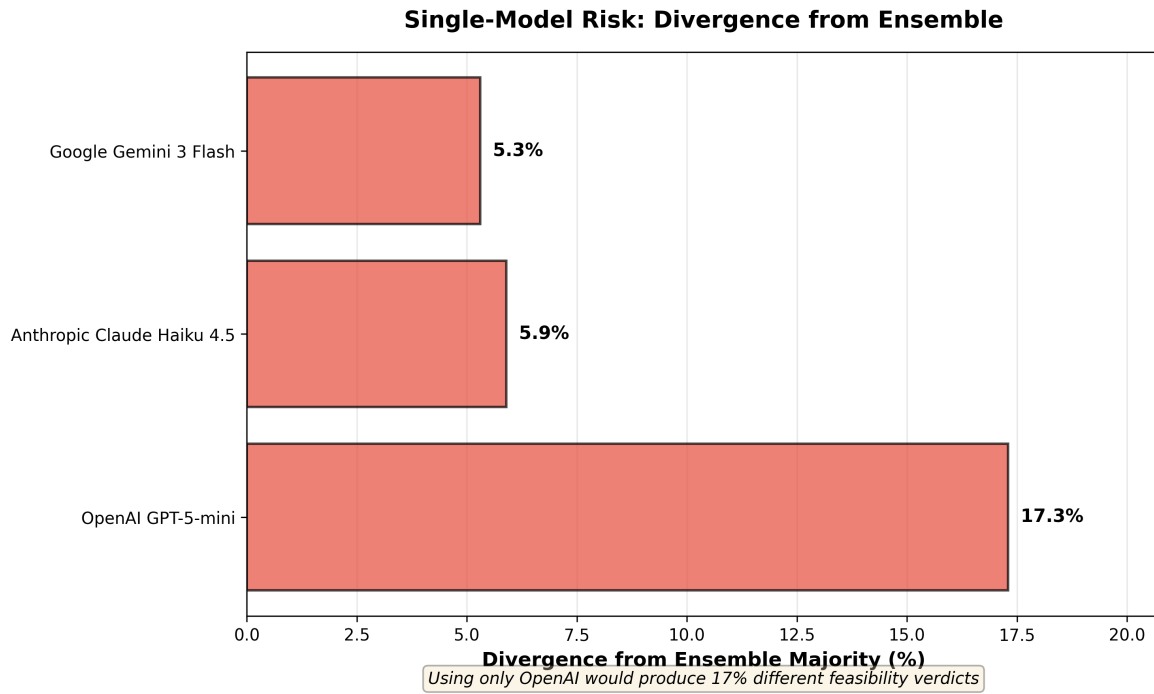


Arbitrator	Synthesis Rate	Style
Google (Gemini 3 Pro)	7%	Deferential — trusts raters
OpenAI (GPT-5.2)	59%	Balanced
Anthropic (Claude Opus 4.5)	77%	Active synthesis

No single model dominates. Ensemble balances conservative and synthetic approaches.

Why Not Just Use One Model?

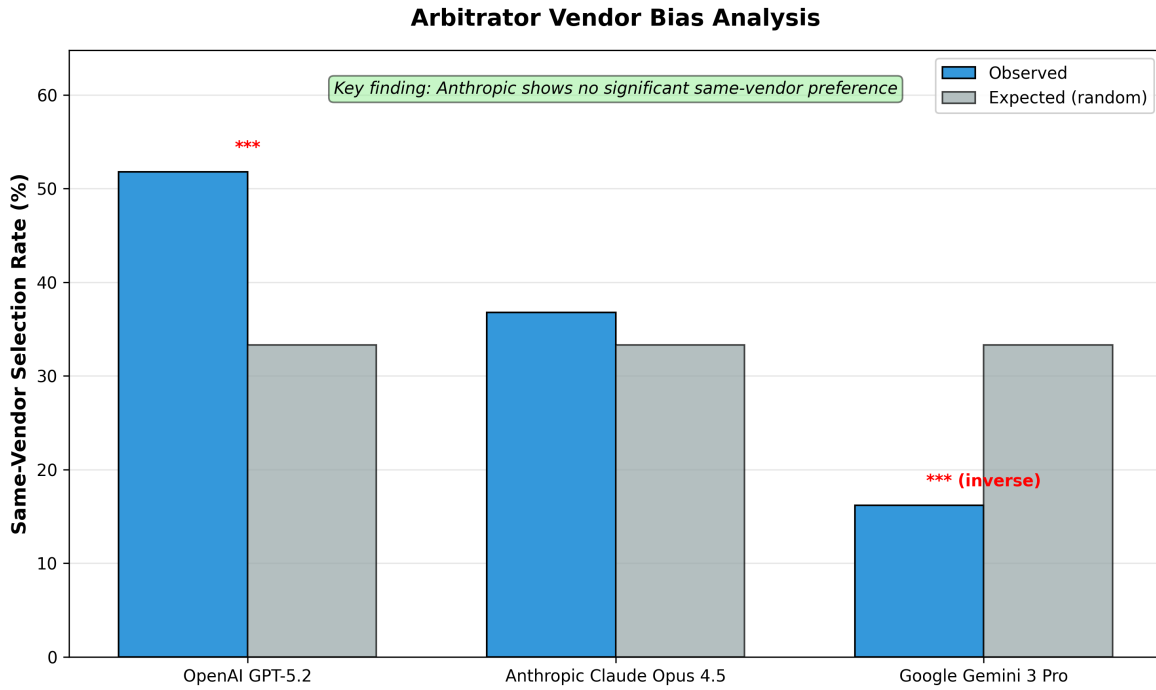
Single-model divergence from ensemble majority:



- **OpenAI alone:** 17.3% different feasibility verdicts (~277 pairs)
- **Anthropic alone:** 5.9% (~94 pairs)
- **Google alone:** 5.3% (~85 pairs)

The 3rd model catches what any 2 would miss.

Do Arbitrators Favor Their Own Vendor's Rater?



Arbitrator	Same-Vendor Selection	Expected	Significant?
OpenAI	51.8%	33.3%	Yes ($p < 0.001$)
Anthropic	36.8%	33.3%	No ($p = 0.159$)
Google	16.2%	33.3%	Yes — AGAINST own

Finding: Anthropic shows no same-vendor preference. Google actively avoids own vendor.

Methodology Validation Summary

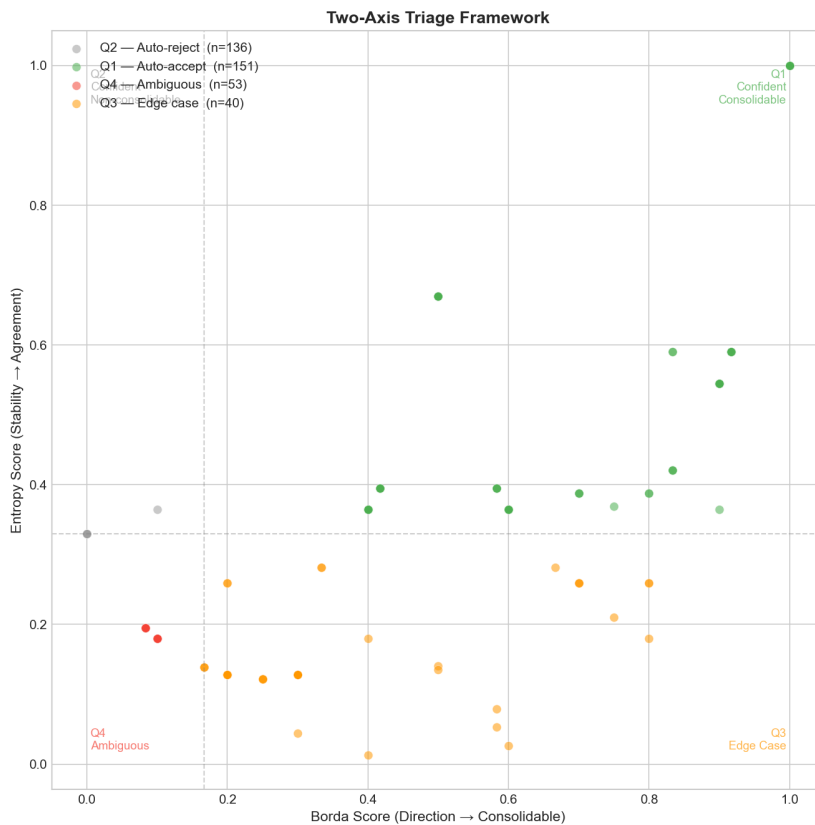
Validation Check	Result	Interpretation
Rater convergence	= 0.611	Substantial — task well-defined
Arbitrator convergence	= 0.843	Almost perfect — disagreements resolvable
Single-model risk	5-17%	Multi-model approach justified

Validation Check	Result	Interpretation
Vendor bias	Mixed	Anthropic unbiased; OpenAI self-favoring

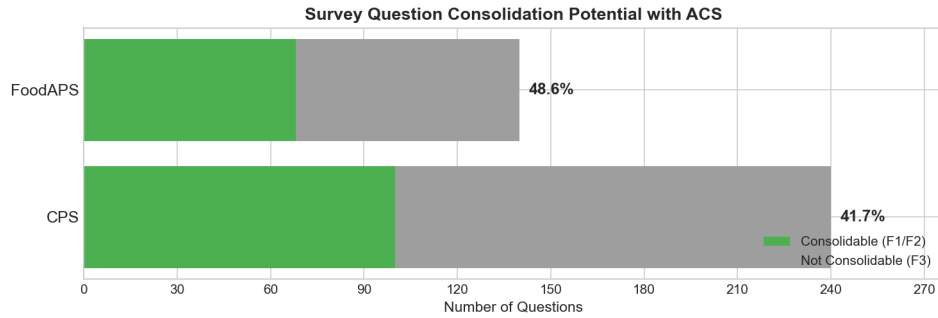
Conclusion: Multi-model ensemble provides robust classifications suitable for survey harmonization analysis.

Question-Level Rollup

- Pair analysis → **380 unique source questions**
- Each with best ACS match
- Two-axis triage: Direction (Borda) × Stability (Entropy)

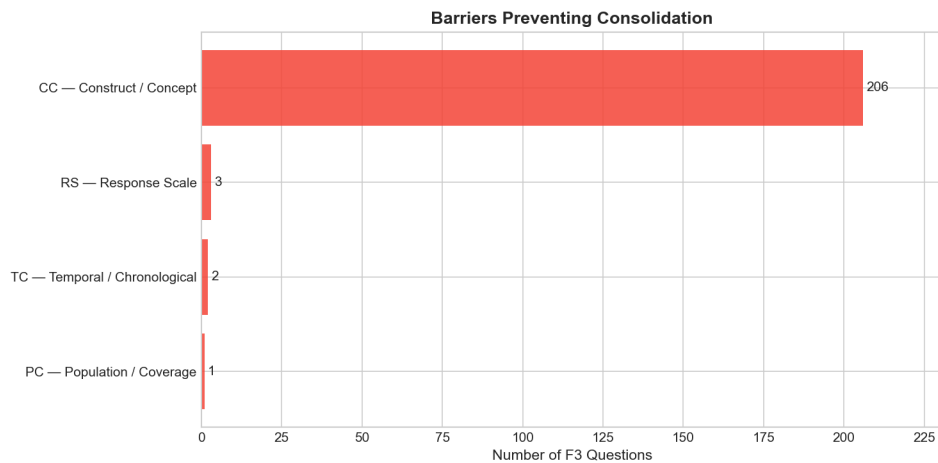


Headline Results



- **CPS:** 41.7% consolidable (100 of 240 questions)
 - **FoodAPS:** 48.6% consolidable (68 of 140 questions)
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Why Questions Can't Consolidate

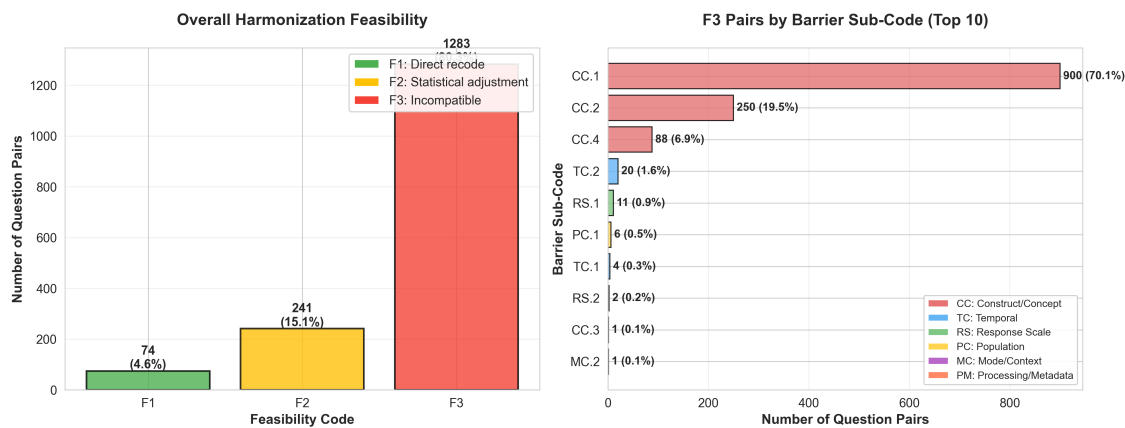


97% of F3 = Construct/Concept differences (CC)

Not fixable barriers — fundamentally different measurements

Harmonization Code Distribution

Harmonization Code Distribution (1,598 Question Pairs)



Left: Overall F1/F2/F3 distribution across 1,598 pairs Right: Top 10 barrier sub-codes for F3 pairs - CC.1 (concept definition) dominates at 70%

Example: High Consolidability (F1)

CPS: “How many hours per week do you USUALLY work?”

ACS: “What is your best estimate of hours per week you usually work?”

- Verdict: **F1** (directly consolidable)
- Confidence: Borda=1.0, Entropy=1.0
- Same construct, same framing, same reference

Example: Medium Consolidability (F2)

CPS: “EXCLUDING overtime, tips, commissions — what is hourly rate?”

ACS: “What is your hourly rate of pay?”

- Verdict: **F2** (consolidable with transformation)
- Transformation needed: scope alignment (ACS includes all components)

Example: Not Consolidable (F3)

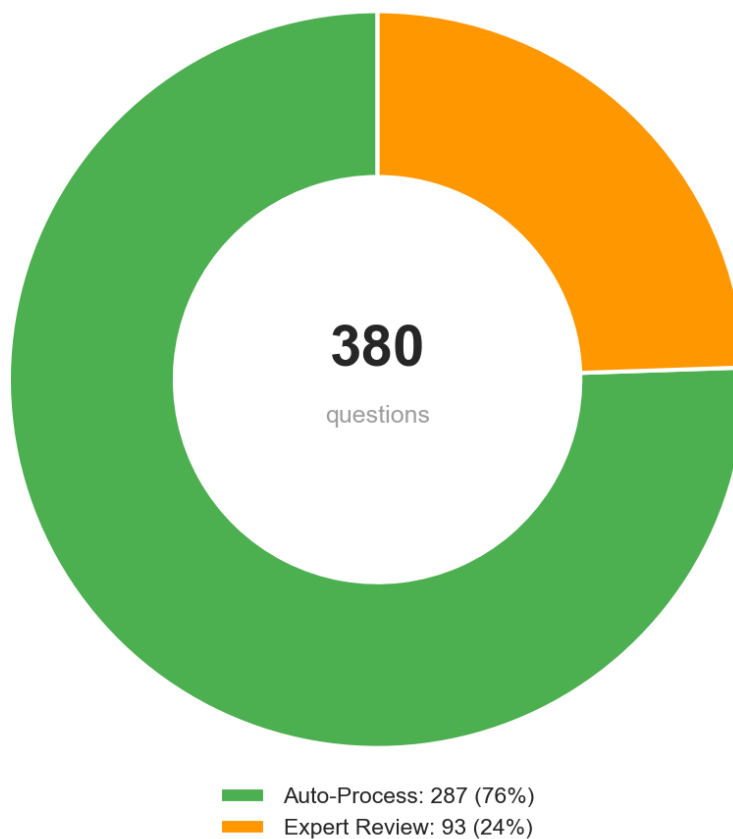
CPS: “Do you WANT to work full time (35+ hours)?”

ACS: “Did this person WORK for pay last week?”

- Verdict: **F3** — Barrier: **CC.1**
 - Same domain (employment), entirely different constructs
 - Preference vs. behavior — not reconcilable
-

Expert Review Load

Classification Confidence Distribution



- **76%** (287 questions) classified with high confidence → spot-check recommended
- **24%** (93 questions) flagged for expert review → **where human judgment adds most value**
- **This IS the efficiency gain:** Experts review 93 questions, not 1,598 pairs

Traditional: Expert reviews all 1,598 pairs = weeks/months AI-assisted: Expert reviews 93 flagged questions = focused effort on highest-ROI decisions

Deliverables

Primary: Expert review table (`expert_review_combined.csv`) - 380 questions with best ACS matches - F1/F2/F3 classifications with confidence scores - Barrier codes and reasoning for each decision - Triage routing (Q1-Q4) for prioritized review

Breakdown: - 168 consolidable (F1+F2) — ready for validation - 212 non-consolidable (F3) — with documented barriers - 93 flagged for expert review — highest-value decisions - Full methodology — reproducible for additional surveys

What This Proves

- **< 1 day processing time** for work that traditionally takes weeks/months
- **Expert efficiency → faster burden reduction** — identify consolidation opportunities at scale
- **Human oversight preserved throughout** — AI provides initial judgment, experts validate
- **The table is the deliverable** — 380 questions, pre-analyzed, ready for expert review

AI doesn't replace burden reduction goals — it accelerates the path to achieving them through expert efficiency

What's Next

1. Expert validation of recommendations
 2. Expand to additional federal surveys
 3. Application to survey editing/imputation
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Summary

Findings: - ~44% **consolidation potential** — 168 questions could reduce respondent burden
- **97% of failures** = construct differences (not fixable)

Method: - < 1 **day processing** for work that traditionally takes weeks/months - **AI provides initial judgment, experts provide final validation** - **The real product:** 380 questions, pre-analyzed, ready for expert review

Impact: Accelerates path to burden reduction through expert efficiency

Expert review table: output/analysis/expert_review_combined.csv Dual value: (1) Identifies burden reduction opportunities (2) Efficiently, preserving expert judgment

Questions?

Contact: [your info]

Repository: [link]

Expert review tables: [link to output]

Appendix

Cost/Quality: Is the 3rd Model Worth It?

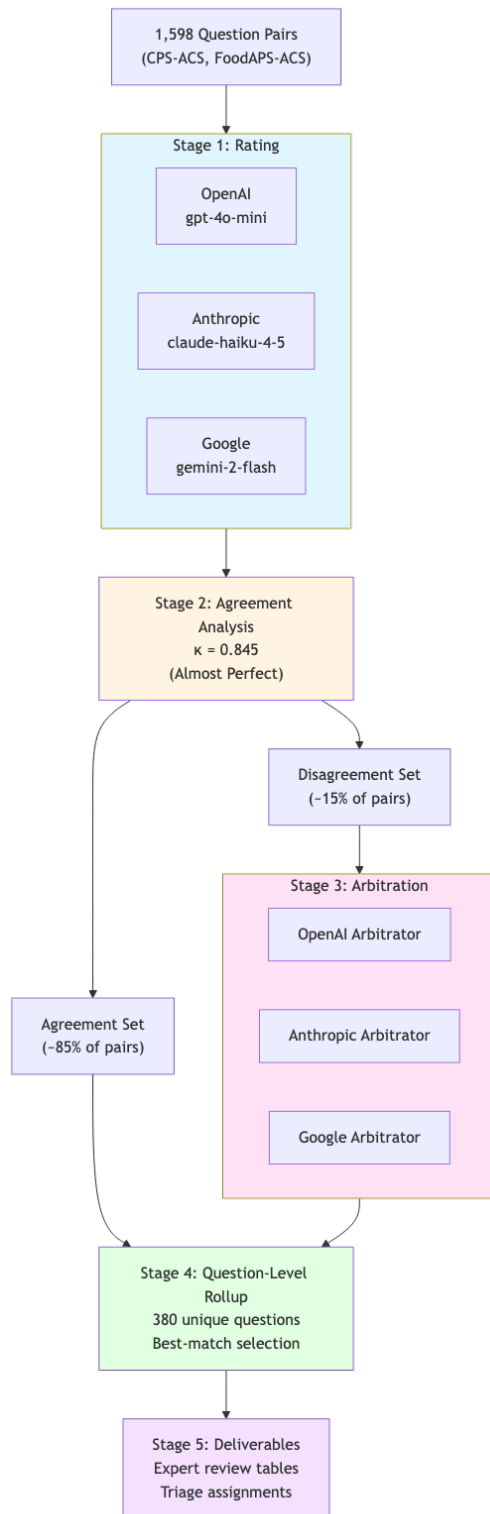
Two-model scenarios — tie frequency:

Pair	Feasibility Ties
OpenAI + Anthropic	23.2%
OpenAI + Google	22.1%
Anthropic + Google	11.0% ← lowest

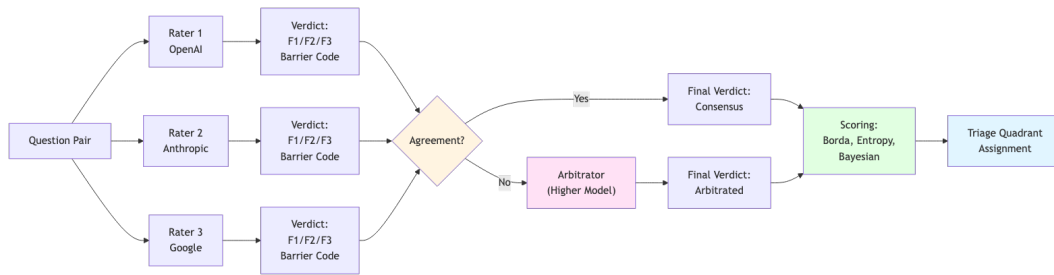
Recommendation: - <1,000 pairs: 2 models + human tiebreaking - >1,000 pairs: 3 models for automated triage

The 3rd model's value is highest when first two disagree.

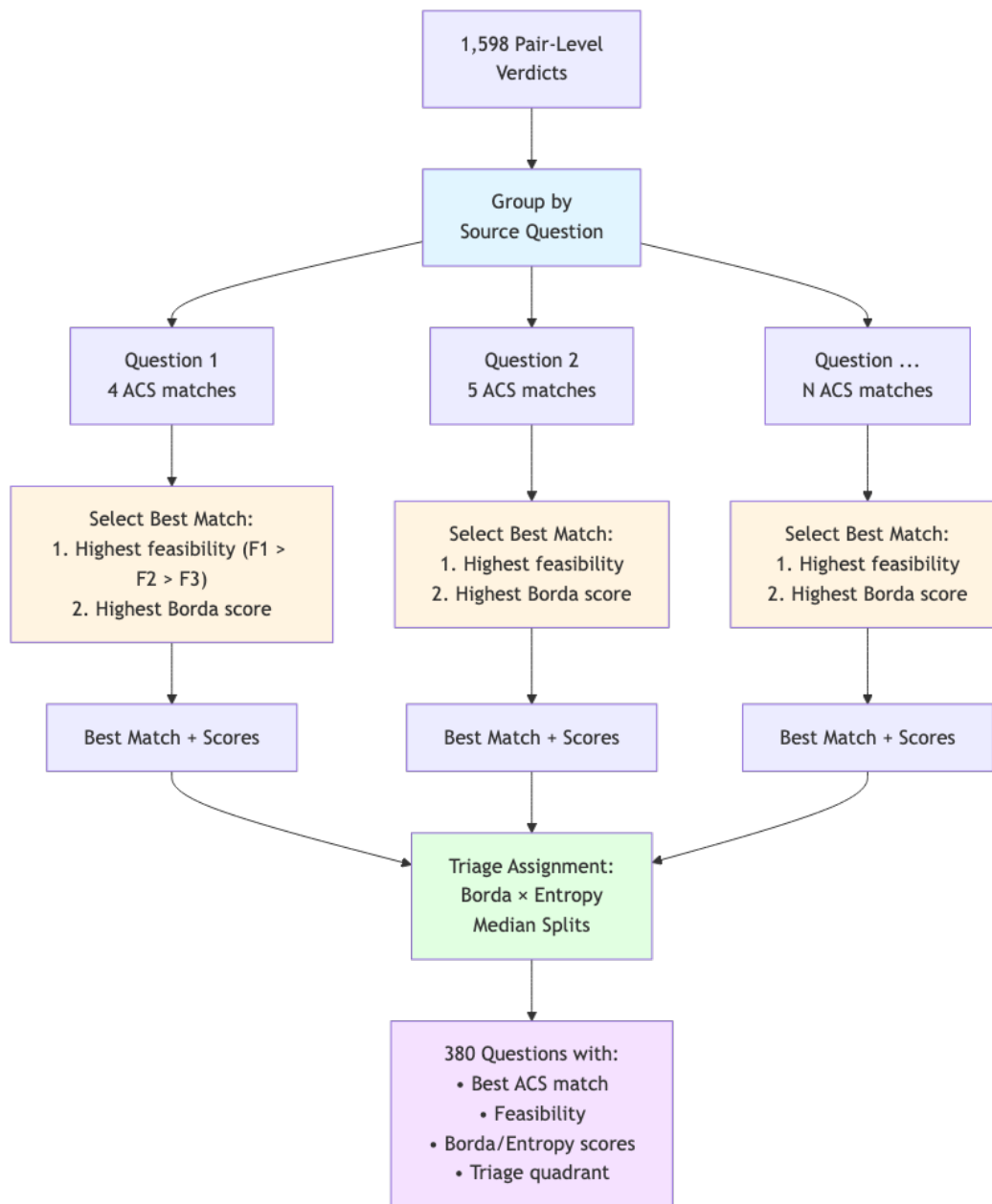
Pipeline Architecture



LLM Ensemble Pattern

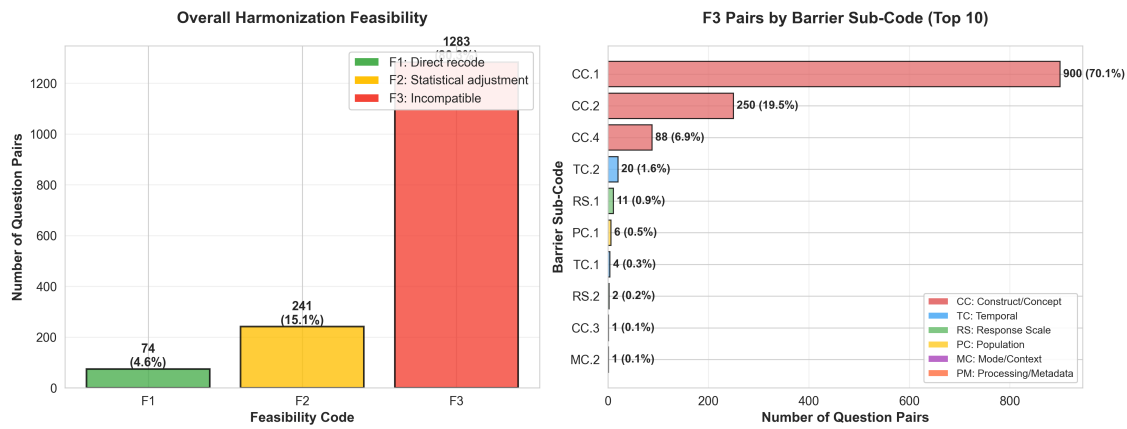


Question-Level Rollup



Harmonization Code Distribution

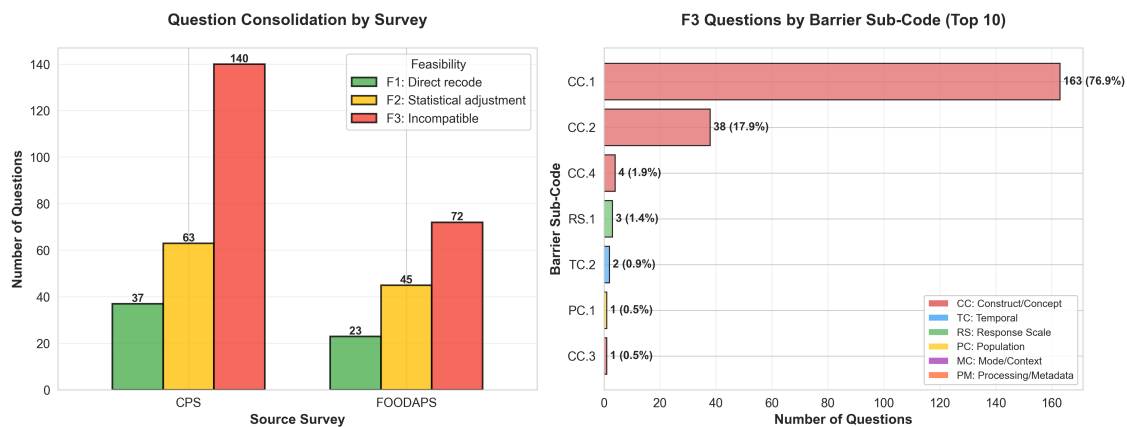
Harmonization Code Distribution (1,598 Question Pairs)



Left: Overall F1/F2/F3 distribution across 1,598 pairs Right: Top 10 barrier sub-codes for F3 pairs - CC.1 (concept definition) dominates at 70%

Question-Level Consolidation Distribution

Question-Level Consolidation Distribution (380 Questions)



Left: CPS (41.7% consolidable) vs FOODAPS (48.6% consolidable) breakdown Right: F3 questions by barrier code - CC.1 dominates at 76.9% of non-consolidable questions

Harmonization Taxonomy

Primary Source: Fortier et al. (2011, 2017) DataSHaPER/Maelstrom framework

Supporting: Wolf et al. (2016), Saris & Gallhofer (2014), Slomczynski & Tomescu-Dubrow (2018)

Code	Feasibility	Definition	Action Required
F1	Direct recode	Mechanically transformable with simple recoding or collapsing	Simple data transformation
F2	Statistical adjustment	Requires modeling, imputation, or bridging studies	Statistical harmonization
F3	Incompatible	Fundamentally different constructs; not harmonizable without re-fielding	No harmonization possible

Fortier et al. (2011) found 38% of variables could be harmonized across 53 studies

Barrier Code Taxonomy (1/3)

Temporal Constraints (TC)

Code	Subtype	Definition	Example
TC	Temporal/Chronological	Reference period or timing differences	—
TC.1	Reference period length	Different recall windows	7-day vs 12-month
TC.2	Temporal framing	Point-in-time vs habitual vs retrospective	“last week” vs “usually”
TC.3	Calendar alignment	Fixed vs rolling reference periods	Calendar year vs “past 12 months”

Construct Constraints (CC)

Code	Subtype	Definition	Example
CC	Construct/Concept	Concept definition or operationalization differences	—
CC.1	Concept definition	Different meaning of core term	“Employment” including/excluding unpaid work
CC.2	Operationalization	Different behavioral indicators	“Food insecurity” via different batteries
CC.3	Boundary conditions	Different thresholds or cutoffs	Income brackets with different boundaries
CC.4	Scope inclusions	Different components counted	“Income” including/excluding specific sources

Barrier Code Taxonomy (2/3)

Population/Coverage Constraints (PC)

Code	Subtype	Definition	Example
PC	Population/Coverage	Universe, frame, or sample design differences	—
PC.1	Universe definition	Target population differs	Civilian vs all residents
PC.2	Frame exclusions	Different sampling exclusions	Group quarters, territories
PC.3	Age bounds	Different age eligibility	15+ vs 16+ vs 18+
PC.4	Geographic scope	Different geographic coverage	50 states vs states + territories

Response Scale Constraints (RS)

Code	Subtype	Definition	Example
RS	Response Scale	Scale type, categories, or format differences	—
RS.1	Scale type	Fundamentally different formats	Binary vs 5-point vs continuous
RS.2	Category structure	Different number/boundaries	5-point vs 7-point Likert
RS.3	Anchoring/labels	Different verbal anchors	“Strongly agree” differences
RS.4	Numeric vs verbal	Number scale vs labeled categories	0-10 slider vs frequency labels

Barrier Code Taxonomy (3/3)

Mode/Context Constraints (MC)

Code	Subtype	Definition	Example
MC	Mode/Context	Interview mode or questionnaire context differences	—
MC.1	Interview mode	Different data collection modes	CATI vs web self-administered
MC.2	Question routing	Different skip patterns	Subset vs all respondents
MC.3	Contextual priming	Preceding questions affect interpretation	Question order effects
MC.4	Proxy response	Proxy vs self-report rules	Household head vs individual

Processing/Metadata Constraints (PM)

Code	Subtype	Definition	Example
PM	Processing/Metadata	Coding, weighting, or documentation differences	—

Code	Subtype	Definition	Example
PM.1	Coding schemes	Different classification systems	Occupation coding differences
PM.2	Derived variables	Different construction algorithms	Poverty calculation methods
PM.3	Documentation gaps	Insufficient metadata	Missing questionnaire context

Triage Quadrants: Operational Framework

Two-axis classification using Borda (direction) and Entropy (stability):

Quadrant	Borda	Entropy	Count	Description	Action
Q1	High	High	151	Confident consolidable — classifiers agreed, leaning accept	Auto-process, spot-check
Q2	Low	High	136	Confident non-consolidable — classifiers agreed, leaning reject	Auto-process, low priority
Q3	High	Low	40	Uncertain accept — leaning yes but contested	Expert review priority
Q4	Low	Low	53	Uncertain reject — genuinely ambiguous	Expert review secondary

Thresholds: Median split on question-level best-match scores

- Borda median: 0.167
- Entropy median: 0.330

Rationale: Separates “what’s the answer?” (Borda) from “how much did they argue?” (Entropy)

Key Citations

Harmonization Framework:

- Fortier, I., et al. (2011). Is rigorous retrospective harmonization possible? Application of the DataSHaPER approach across 53 large studies. *International Journal of Epidemiology*, 40(5), 1314-1328.
- Fortier, I., et al. (2017). Maelstrom Research guidelines for rigorous retrospective data harmonization. *International Journal of Epidemiology*, 46(1), 103-115.

Survey Methodology:

- Wolf, C., et al. (2016). Harmonizing survey questions between cultures and over time. In *The SAGE Handbook of Survey Methodology* (pp. 502-524). SAGE.
- Saris, W. E., & Gallhofer, I. N. (2014). *Design, Evaluation, and Analysis of Questionnaires for Survey Research* (2nd ed.). Wiley.

Data Harmonization:

- Slomczynski, K. M., & Tomescu-Dubrow, I. (2018). Basic principles of survey data recycling. In *Advances in Comparative Survey Methods* (pp. 937-962). Wiley.
- Tomescu-Dubrow, I., et al. (Eds.). (2024). *Survey Data Harmonization in the Social Sciences*. Wiley.